

عبد الرحمن طه

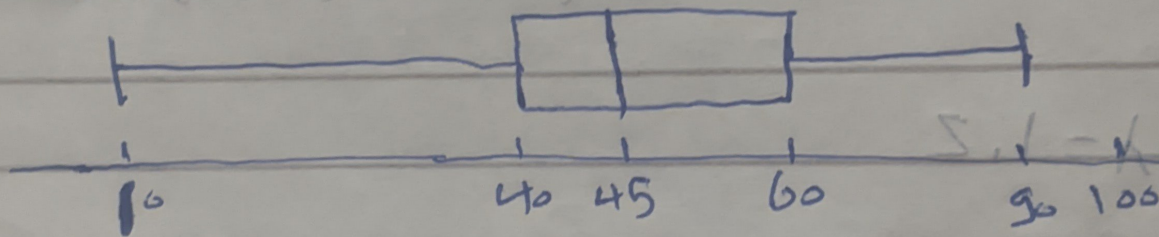
$$Q_1 = 40 \quad Q_2 = 45 \quad Q_3 = 60$$

$$IQR = 20$$

$$\text{upper fence} = 90$$

$$\text{lower fence} = 10$$

①



②

$$Q_2 = 45$$

③

No

Because $15 > 10$

④ right skewed $|Q_2 - Q_1| < |Q_3 - Q_2|$

⑤ coefficient = $\frac{Q_3 - Q_2 - (Q_2 - Q_1)}{Q_3 - Q_1} = \frac{15 - 5}{20} = 0.5$

- conform that distribution is right skewed

⑥ yes it is difficult

عنايه هو positive skewed يعني أغلب التوزيعات الطولية
درجاتهم على الجانب الأيسر التي هو أقل من 45

و 75% من الطلاب درجاتهم أقل من 60

① Discrete distribution table

② In the range 0 to 1

③ 1

④ n (number of trials) and p (probability of success)

⑤ $p < 0.5$ right skewed

$p > 0.5$ left skewed

$p = 0.5$ symmetric

⑥ λ

⑦ mean $= \mu = \sum x p(x) = -0.1$

$$\sigma^2 = E(x^2) - \mu^2 = 4.3 - 0.01 = 4.29$$

$$\sigma = \sqrt{4.29} = 2.07$$

~~⑧~~

8) a) $k = 8$

$$P(X=8) = C_8^8 (0.04)^8 (0.96)^0 = 6.5 \times 10^{-17}$$

b) $k = 2$

$$P(X=2) = C_2^8 (0.04)^2 (0.96)^6 = 0.03$$

c) $k = 0$

$$P(X=0) = C_0^8 (0.04)^0 (0.96)^8 = 0.72$$

9) $\lambda = 1.2$

a) $X = 3$

$$P(X=3) = \frac{e^{-1.2} \cdot 1.2^3}{3 \times 2 \times 1} = 0.286$$

b) $X \geq 2$

$$P(X \geq 2) = 1 - [P(X=0) + P(X=1)]$$

$$P(X=0) = 0.30$$

$$P(X=1) = 0.36$$

$$= 1 - [0.30 + 0.36] = 0.337$$

c) mean = $\lambda = 1.2$, $\sigma = \sqrt{\lambda} = 1.09$